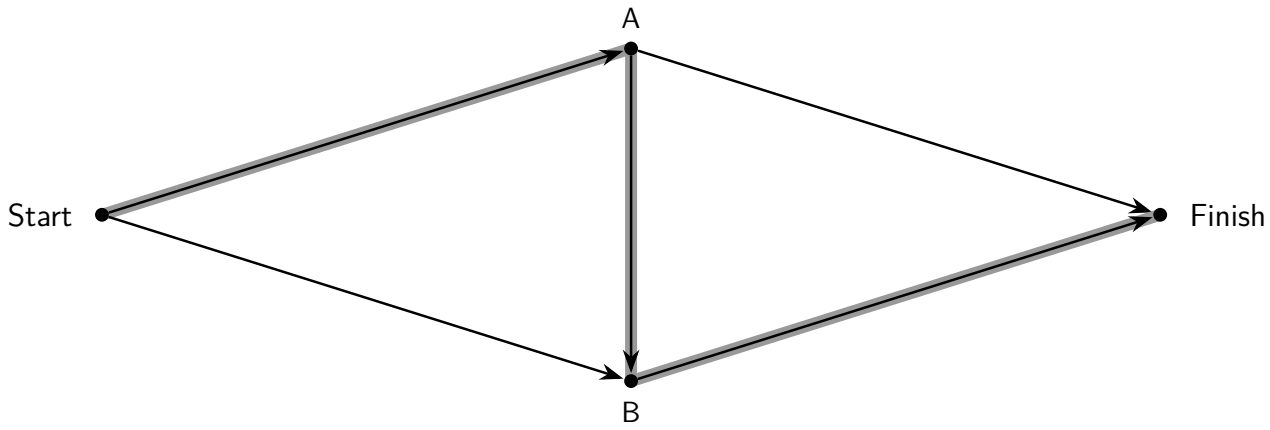
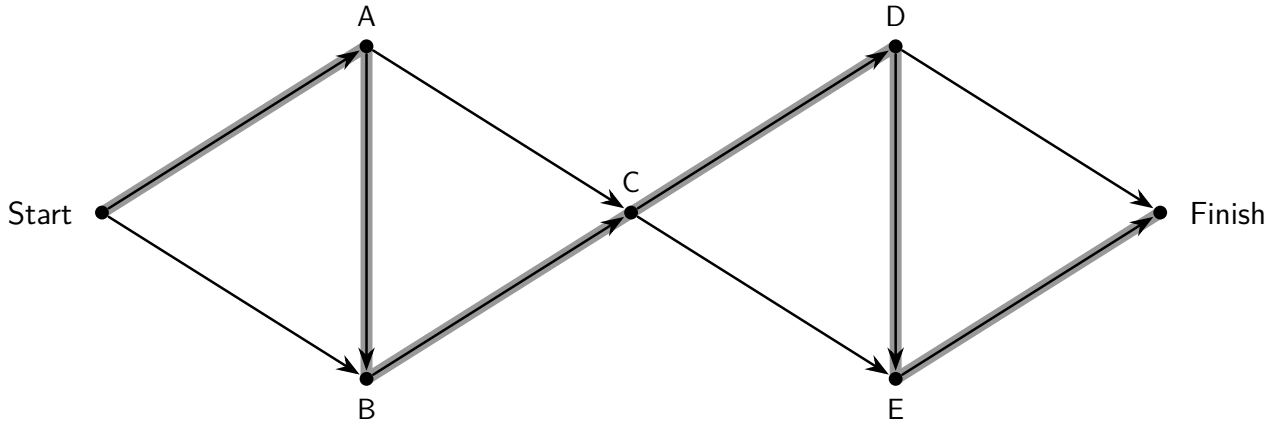
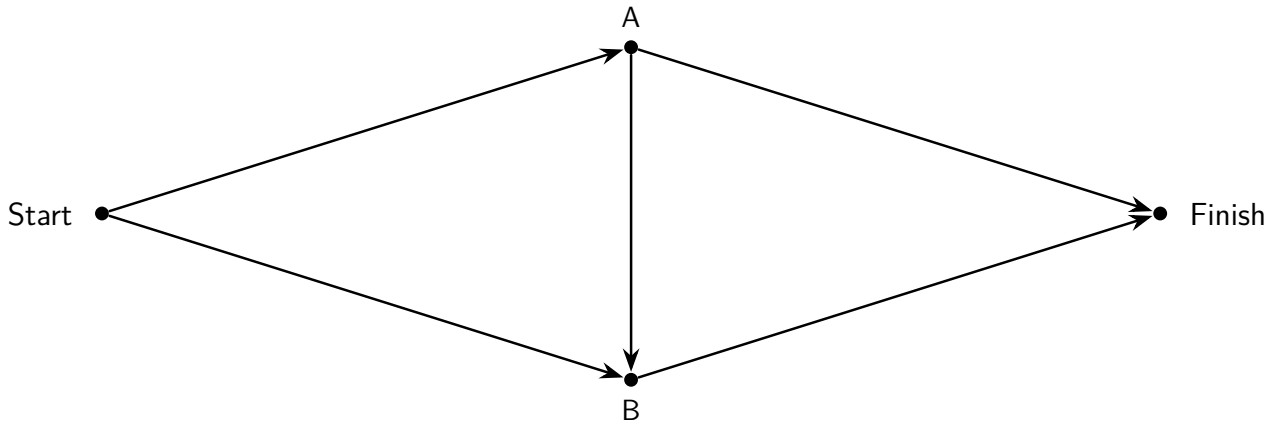
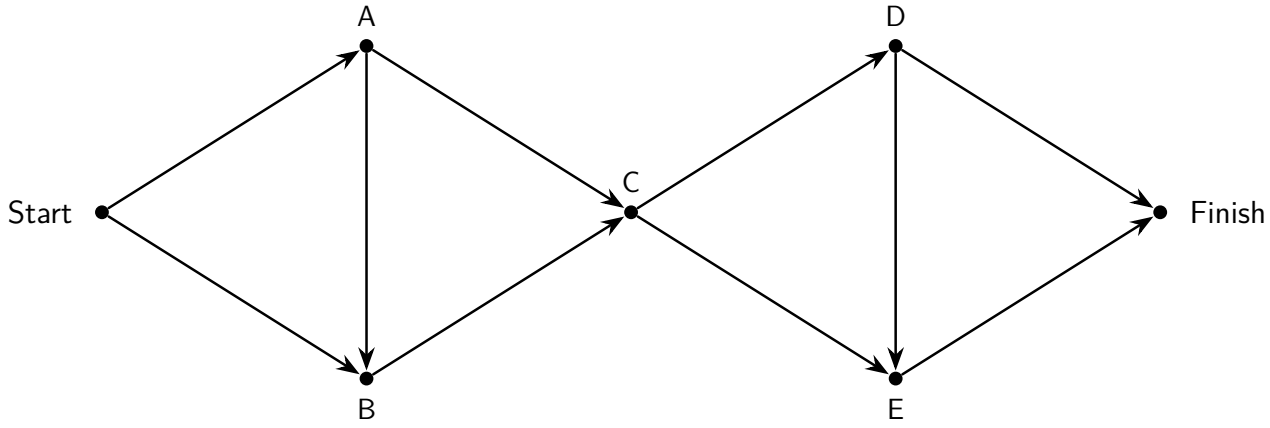
A route follows arrows from Start to Finish without visiting a dot twice. Routes may meet at dots, but each arrow can belong to only one route in a collection. Use a different color or line pattern for each route. A marker closes one arrow.

Problem 1: The thick arrows form reserved routes. No more routes fit beside them. Replace them with a largest collection on each board.

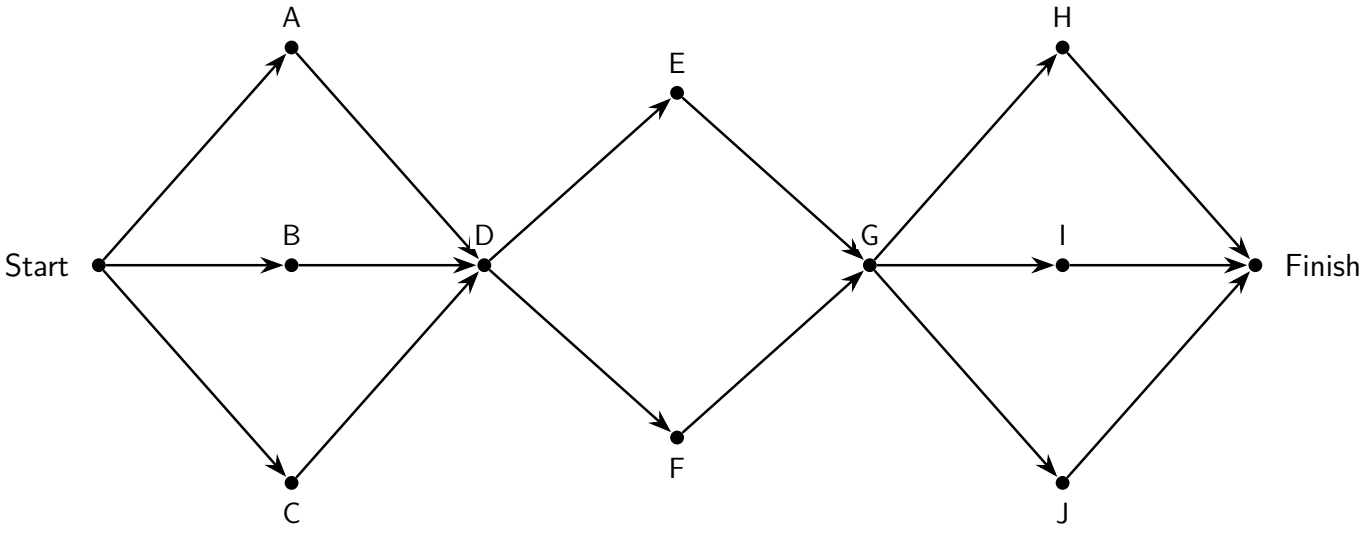





A
Start
Finish
B


A
D
C
Start
Finish
B
E

Problem 2: Find a largest route collection and a smallest set of closing arrows. Explain why your closing arrows stop every route and why the two answers show that your collection is largest.



Problem 3: Split the dots into two sides, with Start on one side and Finish on the other. The arrows pointing from the Start side to the Finish side are called the cut arrows. Find two different splits with exactly two cut arrows and at least one arrow pointing back into the Start side. Explain why closing the cut arrows stops all travel.

